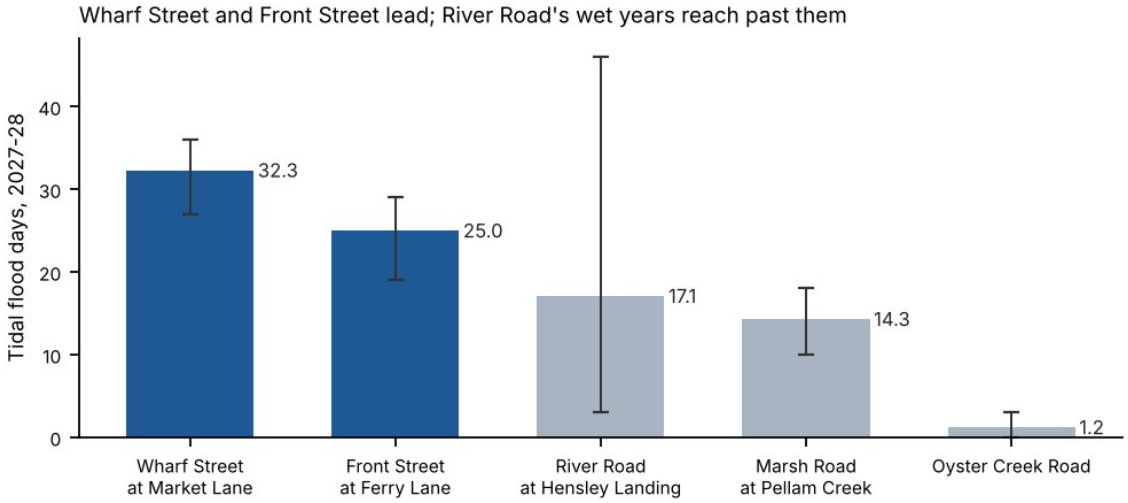
FY2028 tide valves: Wharf Street at Market Lane and Front Street at Ferry Lane

To: Stormwater Advisory Board. From: Cheryl Olsen, PE. 10 November 2026.

Fund the FY2028 valves at Wharf Street at Market Lane (32.3 expected tidal flood days in the 2027-28 tide year) and Front Street at Ferry Lane (25.0). River Road at Hensley Landing (17.1), Marsh Road at Pellam Creek (14.3) and Oyster Creek Road (1.2) go to the FY2029 round. The same two come out on top in 28 of the 36 weather years (in the others, wet late winters on the Brayle, River Road takes Front Street's place) and under all 72 variants of the method we could defend, by at least 1.35 times.

Low point	Flooding elevation (ft NAVD88)	2025-26 days	Expected 2027-28	90% range
Wharf Street at Market Lane	5.13	26	32.3	27 to 36
Front Street at Ferry Lane	5.20	20	25.0	19 to 29
River Road at Hensley Landing	4.95	10	17.1	3 to 46
Marsh Road at Pellam Creek	5.00	17	14.3	10 to 18
Oyster Creek Road	5.03	8	1.2	0 to 3



Harbor gauge. At its 8.6 ft minor flood threshold the gauge expects 27.0 days in 2027-28 (22 to 31 in nine weather years in ten) against 21 observed in 2025-26. The planning sea level, 0.70 ft above the 1983-2001 mean against 0.63 ft in 2025-26, adds 5.8 days; the 2027-28 tides take 0.7 and the weather years against 2025-26's add 1.0, a change of 6.0 (each share averaged over the six orders of applying the three changes; sea level adds 4.1 to 7.0 across them).

Front Street. The lane sits at 5.20 ft NAVD88, the lowest water level the street sensor records on it in readings that persist; the inlet rim is lower but the program counts the lane. The sensor itself recorded 15 of the 20 tidal flood days of 2025-26; the other 5 fell in its September-October outage and come from the harbor's high waters.

Marsh Road and Oyster Creek Road. Pellam Creek's high water follows the harbor's at 1.02 ft per foot until the tide tops the marsh (harbor about 7.43 ft) and at 0.61 above it, so Marsh Road floods at a harbor high water of 8.75 ft, not the 8.54 ft one straight line through the 2024 campaign gives (43 of its 130 high waters stood above the bend). Oyster Creek bends the same way and floods only at 9.11 ft.

River Road. The only campaign ran from January to April 2025 with the Brayle at 7,600 to 69,100 cfs (24,900 on average); Hensley's high waters stand about 0.47 ft higher at 25,000 cfs than at 5,000. Carried day by day on the river's flows, River Road flooded on 10 days in 2025-26; the campaign's average offset carried to the dry autumn king tides would have given 26.

Expected days are averages over the weather and river flows of the tide years 1990-91 to 2025-26, on the gauge's published predictions at the planning sea level (round guidelines, 16 October 2026).